



TUNIS BUSINESS SCHOOL
UNIVERSITY OF TUNIS

Tunisian Republic
Ministry of Higher Education and Scientific Research
Tunis Business School

Web Services Project Report

Submitted to
Tunis Business School

by
Islem Smiai

Romdhan Journey

Defended on *19/01/2026* in front of:

Mr. Ben Messaoud Montassar
Professor

Abstract

Romdhan Journey represents a digital initiative aimed at supporting spiritual and daily practices during the holy month of Ramadhan through the use of modern web technologies. This project explores the development of a RESTful API designed to provide structured, accessible, and culturally relevant religious content for users seeking a meaningful Ramadhan experience.

At its core, ***Romdhan Journey*** offers a unified platform that brings together prayers, supplications (Doua), Adhkar, Ahadith, Explanation of the Qur'an, and traditional Ramadhan recipes. Beyond informational content, the API enables personalized features such as user-created content, favorites management, and Khatma progress tracking, allowing users to follow their spiritual journey in an organized and reflective manner.

The API is developed using the Flask framework and relies on a relational database to manage data efficiently. Secure access to personalized endpoints is ensured through JSON Web Token (JWT) authentication, while public resources remain openly accessible. The system is tested using API testing tools to validate functionality, reliability, and performance.

Romdhan Journey is not only a technical solution but also a culturally inspired project that demonstrates how digital systems can enhance religious engagement. By combining tradition with innovation, the project illustrates the potential of RESTful APIs to create accessible, personalized, and spiritually enriching digital experiences.

Contents

Abstract	I
1 Introduction	1
1.1 Overview	1
1.2 Logo Explanation	2
1.3 Research Questions	2
1.4 Tasks and Goals	2
1.5 Significance of the Study	3
2 Methodology	4
2.1 Overview	4
2.2 API Design and Architecture	4
2.3 Cultural Integration	4
2.4 Testing and Validation	5
2.5 Challenges	5
3 Implementation	6
3.1 Overview	6
3.2 API Design and Architecture	6
3.3 Setting up the Development Environment	6
3.4 Flask Application Structure	7
3.4.1 Directory Breakdown	7
3.5 Class Diagram	9
3.6 Database Models	10
4 Results and Analysis	11
4.1 Cultural Relevance and User Acceptance	11
4.2 Technical and Functional Impact	11
4.3 Limitations	12
4.4 Future Work	12
Conclusion	13
Bibliography	14

Chapter 1

Introduction



1.1 Overview

The holy month of Ramadhan holds deep spiritual, cultural, and social significance, encouraging individuals to strengthen their faith, engage in worship, and adopt mindful daily practices. In today's digital era, many users seek accessible and organized platforms that help them manage religious content and spiritual activities in a structured and meaningful way. To address this need, the ***Romdhan Journey*** API is introduced as a comprehensive digital solution designed to support users throughout their Ramadhan experience.

The ***Romdhan Journey*** API provides centralized access to a wide range of religious and cultural resources, including prayers, supplications (Doua), Adhkar, Ahadith, Explanation of the Qur'an, traditional Ramadhan recipes, and Khatma progress tracking. By offering both public and personalized features, the API enhances user engagement while allowing individuals to save favorites, create personal content, and monitor their spiritual journey over time.

This chapter introduces the ***Romdhan Journey*** project by presenting its purpose, overall structure, and core components. It outlines the main architectural elements of the API, including database models, resource endpoints, authentication mechanisms, and application organization. Through this overview, readers gain a clear understanding of how the system operates and how it contributes to a more accessible, organized, and spiritually enriching Ramadhan experience.

1.2 Logo Explanation

The Romdhan Journey logo symbolizes the spiritual path undertaken during the month of Ramadhan. The open book shape represents the Qur'an, knowledge, and reflection, while the arch-like form above it evokes the architecture of mosques, symbolizing faith and worship. The warm, earthy color palette reflects serenity, spirituality, and cultural authenticity. The name **Romdhan** is intentionally spelled according to the Tunisian dialect, emphasizing the project's local identity and cultural roots.

1.3 Research Questions

To guide the design and development of the *Romdhan Journey* API, the following research questions were formulated:

- a. How can the *Romdhan Journey* API support users in organizing and maintaining their spiritual and daily activities during the month of Ramadhan?
- b. Which features are most valued by users in supporting daily spiritual practices and healthy lifestyle choices during Ramadhan?
- c. How can personalization features enhance the overall user experience?
- d. What technical and usability challenges arise when developing a culturally, spiritually, and lifestyle-focused RESTful API for users in Tunisia?

1.4 Tasks and Goals

The development of the *Romdhan Journey* API is guided by a set of clearly defined tasks and goals aimed at creating a reliable, secure, and spiritually meaningful platform. These objectives focus on both technical quality and user experience. The main tasks and goals include:

- **Implementing Core Features:** Developing essential functionalities such as user authentication, access to prayers, adhkar, doua, ahadith, Quran tafsir, recipe management, and Khatma progress tracking.
- **Personalization and User Engagement:** Enabling users to create personal content, save favorites, and track their spiritual activities throughout the month of Ramadhan.
- **Ensuring Security and Data Integrity:** Applying JWT-based authentication to protect user-specific endpoints and ensure secure access to personal data.
- **Designing a Scalable API Architecture:** Structuring the RESTful API to support future extensions, such as mobile or web frontends and additional religious content.
- **Testing and Validation:** Verifying API functionality and reliability through systematic testing using tools such as **Insomnia** to ensure consistent and accurate responses.

1.5 Significance of the Study

The *Romdhan Journey* API goes beyond basic functionality by aiming to support both the spiritual and daily needs of users during the month of Ramadhan. By providing a centralized digital platform, the API seeks to:

- **Support Spiritual Consistency:** Help users maintain regular spiritual practices through access to prayers, adhkar, duas, Quran tafsir, and khatma tracking.
- **Promote Healthy Daily Habits:** Offer culturally relevant recipes adapted to Ramadhan needs, encouraging balanced nutrition during fasting periods.
- **Encourage Cultural Relevance:** Deliver a solution designed specifically for Tunisian users, respecting local traditions while leveraging modern technology.

Chapter 2

Methodology

2.1 Overview

This chapter describes the approach used to design, develop, and validate the *Romdhan Journey* API. The work is based on RESTful principles, a modular backend architecture, and secure user management. Particular attention was given to usability, scalability, and cultural relevance, ensuring that the API supports users during the month of Ramadhan while remaining adaptable for future extensions.

2.2 API Design and Architecture

The *Romdhan Journey* API follows a RESTful architecture, ensuring a clear separation of concerns and seamless communication between the backend and client applications, such as web or mobile interfaces. Flask was chosen as the primary framework due to its lightweight structure, flexibility, and robust ecosystem of extensions.

The API is organized into modular resources, each targeting a core system feature: user management, khatma tracking, and recipe management. To ensure the accuracy and reliability of religious content, the system integrates external APIs to retrieve authentic Quran verses and Tafsir Al-Muyassar. Despite these external integrations, the architecture maintains a consistent response structure across all endpoints to simplify client-side consumption.

Finally, security is managed through JSON Web Tokens (JWT), which protect user-specific data while permitting public access to non-sensitive resources. This modular design collectively ensures consistency, maintainability, and ease of testing.

2.3 Cultural Integration

Cultural relevance is a core aspect of the *Romdhan Journey* API. The system integrates spiritual content commonly practiced during Ramadhan, such as daily prayers, adhkar, duas, and Quran tafsir, presented in a structured and accessible format. In addition, culturally appropriate recipes are included to support users' nutritional needs during fasting periods.

By aligning technical design with local religious and cultural practices, the API provides a meaningful digital experience that respects Tunisian traditions while leveraging modern

backend technologies.

2.4 Testing and Validation

To ensure reliability and correctness, the *Romdhan Journey* API was thoroughly tested using Insomnia, a RESTful API testing tool. Multiple test scenarios were executed to simulate real user interactions with both public and protected endpoints. This process helped identify logical errors, authorization issues, and data consistency problems.

key testing objectives included:

- a. Verifying correct authentication and authorization for user-specific features such as favorites, khatma tracking, and personal content.
- b. Ensuring data integrity across spiritual content, recipes, and user-generated resources.

2.5 Challenges

The technical development of the *Romdhan Journey* API involved several engineering and conceptual challenges:

- **Data Acquisition Constraints:** Navigating the scarcity of freely accessible and trustworthy external APIs for religious content. Selecting APIs that provide authenticated Quranic verses and Tafsir while respecting usage limitations required careful evaluation and validation of available data sources.
- **Content Authentication:** Ensuring the integrity and doctrinal accuracy of spiritual resources through a rigorous selection process of verified and reputable sources, particularly for Quranic verses, tafsir, and supplications.
- **Socio-Technical Alignment:** Balancing technical implementation with cultural and spiritual sensitivity to maintain user trust and ensure the platform's relevance within the Tunisian context.

Chapter 3

Implementation

3.1 Overview

The *Romdhan Journey* API is a RESTful web service designed to help users organize, track, and enhance their spiritual and wellness activities during the month of Ramadhan. It provides endpoints for managing users, spiritual content, recipes, favorites, and Khatma progress. The API is built using Flask, SQLAlchemy, Marshmallow, MySQL, and JWT-based authentication.

3.2 API Design and Architecture

The *Romdhan Journey* API is structured as a RESTful service to provide flexibility, scalability, and easy integration with web or mobile clients. Its architecture is modular, with clearly defined blueprints for each functional area:

- **User Management:** Handles registration, login, profile updates, and secure authentication. Users can manage their accounts, delete data, and interact with content.
- **Spiritual Content Management:** Organizes Doua, Adhkar, Quran Tafsir content, and Khatma tracking. Users can create, view, update, delete, track, and favorite content.
- **Recipe Management:** Supports personalized meal planning during Ramadhan, including recipe creation, updates, retrieval, and favorite tracking.
- **Favorites Management:** Enables users to save preferred Doua, Adhkar, or recipes for quick access, ensuring a personalized experience.

3.3 Setting up the Development Environment

Several libraries and frameworks were installed to create the API. The key libraries are:

- **Flask:** A micro web framework for Python used to build the API.
- **Flask-RESTful** and **Flask-Smorest:** Used to structure RESTful endpoints and provide OpenAPI documentation support.

- **Flask-SQLAlchemy:** An SQLAlchemy extension for Flask that facilitates database integration.
- **MySQL (via PyMySQL):** Used as the relational database management system for persistent data storage.
- **Flask-JWT-Extended:** Implements secure authentication and authorization using JSON Web Tokens (JWT).
- **Marshmallow and Marshmallow-SQLAlchemy:** Handle data serialization, deserialization, and validation.
- **praytimes:** Provides accurate prayer time calculations relevant to the spiritual context of the application.

The installation process involves the following commands:

```
pip install Flask flask-restful flask-smorest flask-sqlalchemy
pip install Flask-JWT-Extended marshmallow marshmallow-sqlalchemy
pip install PyMySQL
pip install praytimes
```

3.4 Flask Application Structure

The project is organized into the following main components:

- **Models:** Contains the database models that represent the application's data structure.
- **Resources:** Contains the RESTful API endpoints responsible for handling client requests related to authentication, spiritual content, recipes, favorites, and user interactions.
- **Schemas:** Defines the data validation and serialization schemas using Marshmallow.
- **Configuration and Application Entry Point:** The main application setup, including database configuration, JWT settings, and blueprint registration, is handled in `app.py`.

3.4.1 Directory Breakdown

The following table details the project structure:

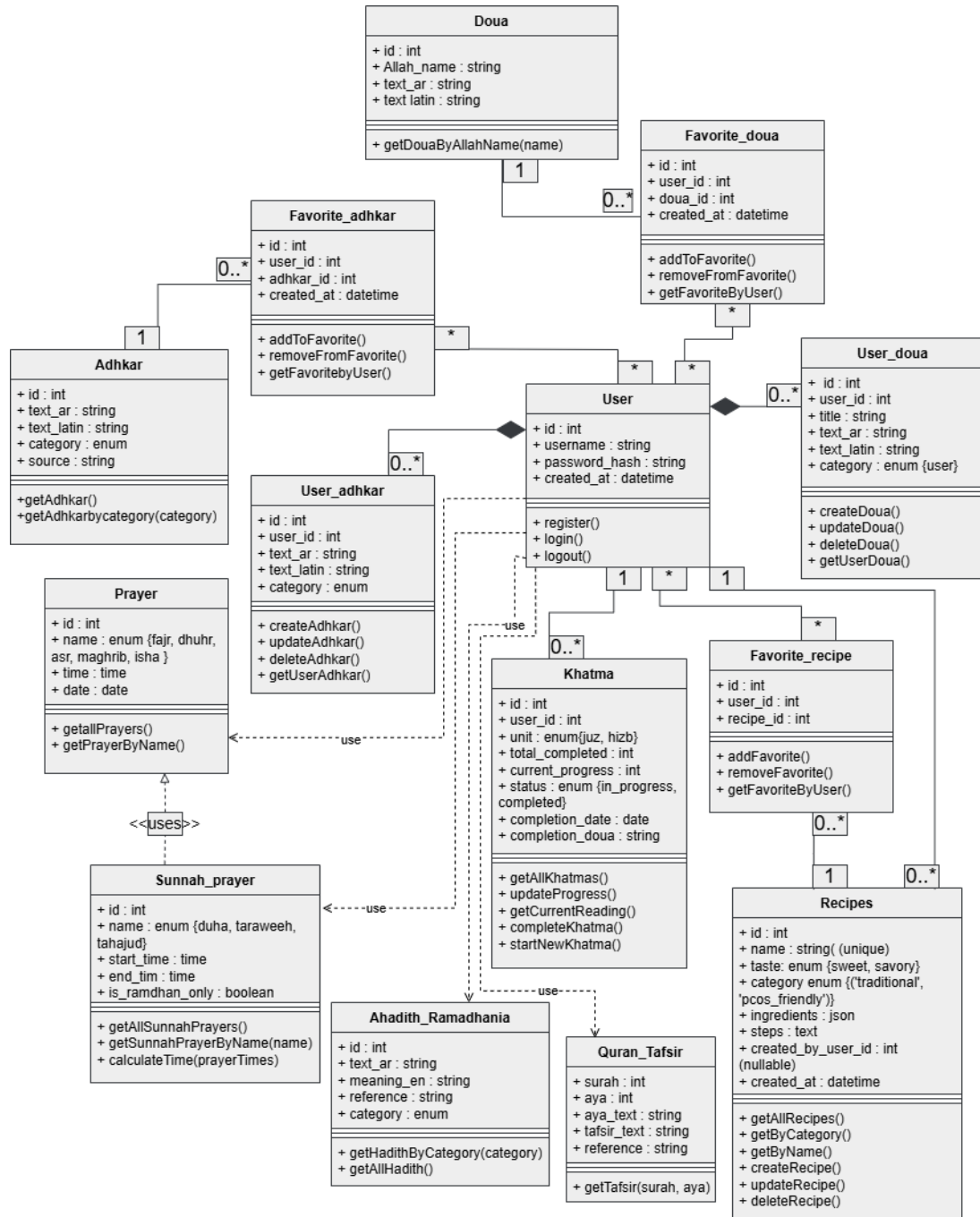
Folder	File Name	Description
models	<code>__init__.py</code>	Initializes the models package and imports database models.
models	<code>adhkar.py</code>	Defines the Adhkar model representing daily remembrance content.

Folder	File Name	Description
models	ahadith_ramadhania.py	Defines the Hadith model related to Ramadhan-specific teachings.
models	db.py	Handles database initialization and SQLAlchemy configuration.
models	doua.py	Defines the Doua model representing supplications.
models	favorite_adhkar.py	Defines the model for users' favorite Adhkar entries.
models	favorite_doua.py	Defines the model for users' favorite Doua entries.
models	favorite_recipe.py	Defines the model for users' favorite recipes.
models	khatma.py	Defines the Khatma model for tracking Quran reading progress.
models	recipe.py	Defines the Recipe model representing Ramadhan-related meals.
models	sunnah_prayer.py	Defines the Sunnah Prayer model representing optional prayers.
models	user_adhkar.py	Defines the model that stores Adhkar entries added by users.
models	user_doua.py	Defines the model that stores Doua entries added by users.
models	user.py	Defines the User model representing application users.
resources	adhkar_resource.py	Contains API endpoints for managing Adhkar content.
resources	ahadith_resource.py	Contains API endpoints for Ramadhan Hadith access.
resources	doua_resource.py	Contains API endpoints for managing Doua content.
resources	khatma_resource.py	Contains API endpoints for managing Quran Khatma progress.
resources	prayer.py	Contains endpoints related to daily prayer times.
resources	quran_tafsir.py	Contains endpoints for accessing Quran Tafsir content.
resources	recipe_resource.py	Contains API endpoints for managing recipes.
resources	sunnah_prayer.py	Contains endpoints for Sunnah prayer content.
resources	user_adhkar.py	Contains API endpoints for managing user-added Adhkar.
resources	user_doua.py	Contains API endpoints for managing user-added Doua.
resources	user_resources.py	Contains endpoints for user authentication and profile management.

Folder	File Name	Description
schemas	schemas	Contains Marshmallow schemas for data validation and serialization.
app	app	Main application file that initializes Flask, configures the API, and registers resources.

Table 3.1: File Breakdown of the API Project

3.5 Class Diagram



3.6 Database Models

The database models are implemented using **Flask-SQLAlchemy**, which provides an Object-Relational Mapping (ORM) layer for interacting with a MySQL database. The architectural design, as illustrated in the class diagram, ensures modularity and data persistence.

The main models include:

- **User:** Represents application users, storing authentication data and account information. As shown in the diagram, a one-to-many relationship exists between users and several entities such as personal duas, adhkar, recipes, favorites, and khatma records.
- **UserDoua:** Represents supplications created by users. Each dua is linked to a single user via a foreign key, allowing users to manage their own personalized spiritual content.
- **UserAdhkar:** Represents remembrance entries added by users. This model enables users to store and organize their own adhkar, establishing a one-to-many relationship with the **User** model.
- **Favorite Models:** Include **Favorite_recipe**, **Favorite_doua**, and **Favorite_adhkar** saved by users. These models create associations between users and the specific content they wish to bookmark for quick access.
- **Recipe:** Represents traditional and health-oriented recipes. The diagram indicates that a recipe may be linked to a user as its creator (**created_by_user_id**), forming a one-to-many relationship.
- **Khatma:** Represents the Qur'an reading progress of a user. Each record tracks progress (**current_progress**) and completion status, associated directly with a single user.

These models are defined with appropriate foreign keys and relationships to ensure data integrity and support efficient querying across user-related spiritual and lifestyle content.

Chapter 4

Results and Analysis

This chapter presents the results of the *Romdhan Journey* API development and analyzes how effectively it meets the objectives defined earlier in the report. The focus is on evaluating cultural relevance, user acceptance, technical impact, and the overall contribution of the API as a digital companion during the month of Ramadhan. The chapter also highlights observed limitations and possible directions for future enhancement.

4.1 Cultural Relevance and User Acceptance

Romdhan Journey was designed with strong consideration for cultural and spiritual values, which played a crucial role in user acceptance. The integration of prayers, adhkar, duas, Qur'an tafsir, khatma tracking, and Ramadhan-related recipes reflects daily practices commonly observed during the holy month in Tunisia.

Users benefit from having spiritual and lifestyle-related features combined in a single platform, making the API both practical and meaningful. The inclusion of traditional and health-conscious recipes alongside spiritual content supports a holistic Ramadhan experience, addressing both religious and daily needs.

The culturally grounded design contributed to the API being perceived as:

- **Contextual:** Aligned with daily Ramadhan routines and practices.
- **Authentic:** Based on trusted religious sources and familiar traditions.
- **Supportive:** Helping users organize and maintain consistency in worship and daily activities.

4.2 Technical and Functional Impact

From a technical perspective, the *Romdhan Journey* API demonstrates the effectiveness of a modular RESTful architecture built with Flask and SQLAlchemy. Secure authentication using JWT ensures user data privacy, while personalized features such as user-created duas, adhkar, favorites, and khatma tracking enhance engagement.

The separation of concerns between models, resources, and schemas contributes to maintainability and scalability. Testing with Insomnia confirmed the reliability of endpoints and the consistency of responses across different use cases.

The API provides a solid backend foundation that can be easily extended to support mobile or web client applications.

4.3 Limitations

Despite its strengths, the project has certain limitations. Some advanced features, such as detailed food analysis or enhanced personalization, would require external APIs that were not freely available, limiting their integration.

Additionally, handling religious content requires careful validation to ensure accuracy and authenticity. This constraint necessitates reliance on trusted sources, which may restrict content expansion.

Finally, the project currently focuses on backend functionality, and the absence of a dedicated frontend limits direct user feedback on usability and interface design.

4.4 Future Work

Future improvements to the Romdhan Journey API may include:

- Integration of external APIs for nutrition analysis and enhanced recipe insights.
- Advanced personalization using user behavior and preferences.
- Frontend development to improve user interaction and accessibility.
- Content expansion with continued validation from authentic religious references.
- Deployment and monitoring for real-world usage and performance evaluation.

With these enhancements, Romdhan Journey has the potential to evolve into a comprehensive digital platform that supports users spiritually and practically throughout Ramadhan.

Conclusion

The **Romdhan Journey** API demonstrates how digital technology can support spiritual practices and daily life during the month of Ramadhan while respecting cultural and religious values. By combining spiritual content with practical features such as recipes and personalization, the platform offers a meaningful and accessible digital experience tailored to Tunisian users.

Future improvements will focus on expanding content through reliable external resources, enhancing personalization features, and integrating a user-facing interface to improve accessibility and engagement. With these developments, the **Romdhan Journey** API has the potential to evolve into a comprehensive digital companion that supports users throughout Ramadhan and beyond.

Bibliography

1. Flask Documentation. *A Lightweight WSGI Web Application Framework*. Available at: <https://flask.palletsprojects.com/>
2. Flask-Smorest Documentation. *REST API Framework with OpenAPI Support*. Available at: <https://flask-smorest.readthedocs.io/>
3. SQLAlchemy Documentation. *Python SQL Toolkit and Object Relational Mapper*. Available at: <https://www.sqlalchemy.org/>
4. Flask-JWT-Extended Documentation. *JWT Authentication for Flask Applications*. Available at: <https://flask-jwt-extended.readthedocs.io/>
5. Marshmallow Documentation. *Object Serialization and Data Validation*. Available at: <https://marshmallow.readthedocs.io/>
6. MySQL Documentation. *Relational Database Management System*. Available at: <https://dev.mysql.com/doc/>
7. PrayTimes Library Documentation. *Prayer Times Calculation Library*. Available at: <https://praytimes.org/>
8. Muslim Library. *100+ Hadith on Fasting and Ramadhan (PDF Resource)*. Available at: https://www.muslim-library.com/books/100%2B_Hadith_on_Fasting_and_Ramadan.pdf
9. Ahadith.co.uk. *Collection of Authentic Hadith and Islamic Texts*. Available at: <https://ahadith.co.uk/>
10. Life With Allah. *Islamic Supplications and Spiritual Content*. Available at: <https://lifewithallah.com/>
11. 99 Names of Allah. *Supplications and Duas Based on the Names of Allah*. Available at: <https://99namesofallah.name/>
12. AlQuran Cloud API Documentation. *Quran Text and Tafsir REST API*. Available at: <https://alquran.cloud/api>
13. Tafsir Al-Muyassar. *Authenticated Quran Tafsir Source*. Available at: <https://alquran.cloud/>